MILITARY SPECIFICATION SHEET

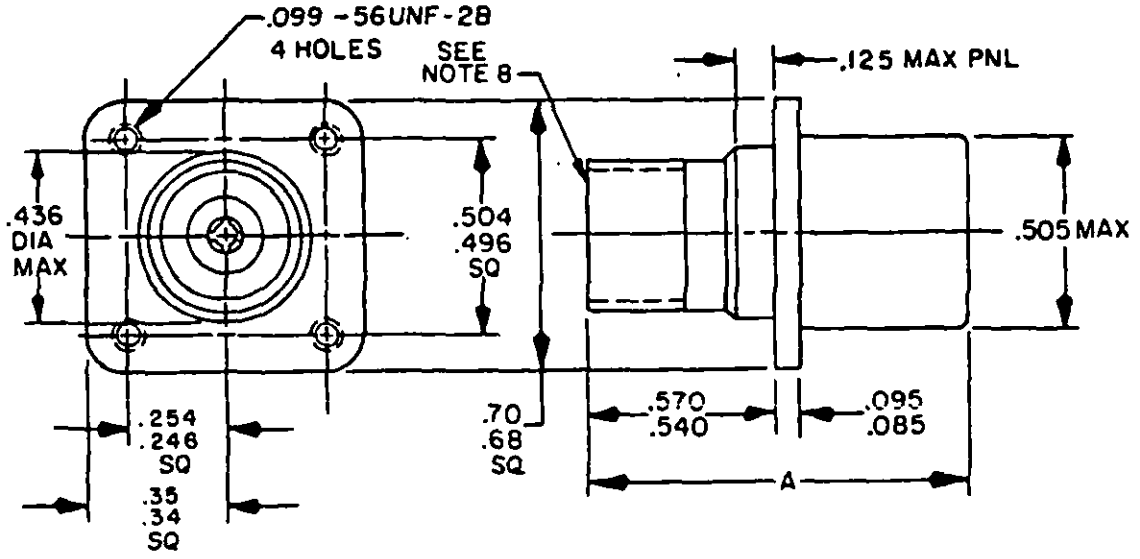
CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES TNC (CABLED), SOCKET CONTACT, FLANGE MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE FOR CATEGORY B, SEE TABLE IV



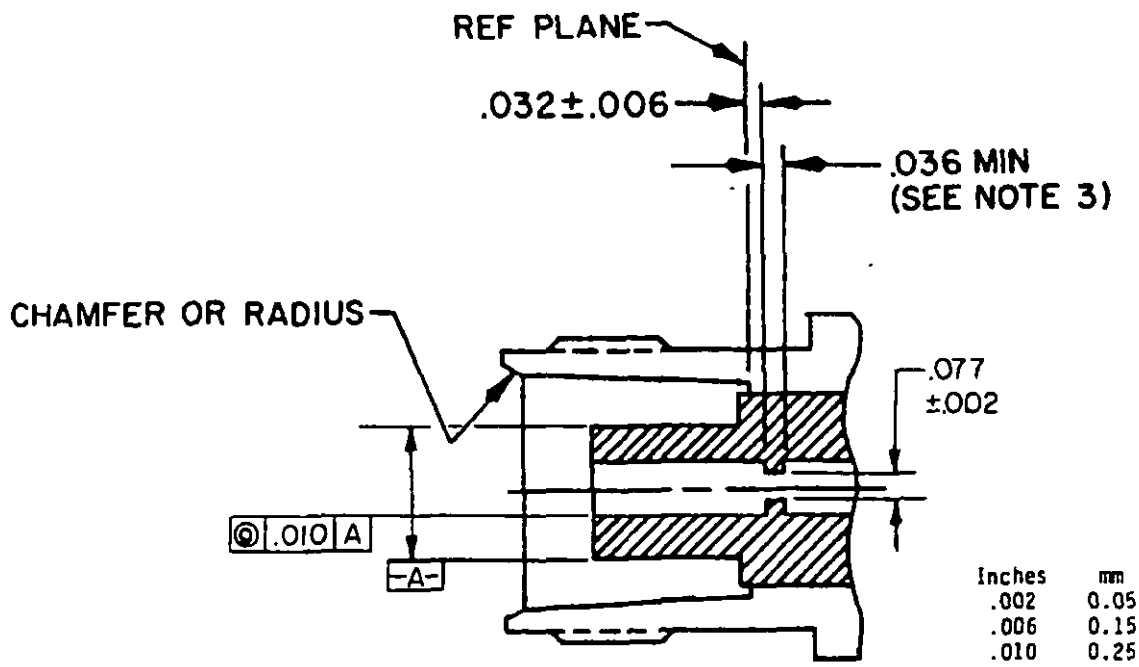
Inches	mm	Inches	mm
.085	2.16	.436	11.07
.095	2.41	.496	12.60
.099	2.59	.504	12.80
.125	3.18	.505	12.83
.246	6.25	.540	13.72
.254	6.45	.570	14.48
.34	8.64	.68	17.27
.35	8.89	.70	17.78

NOTES:
1. Dimensions are in inches.
2. For dimensions A see table I. and III.
3. Metric equivalents are given for general information only.
4. Receptacle not recommended for use on panels with greater than .078 (1.98 mm) maximum thickness.
5. Wrench flats are to accommodate standard wrench opening in accordance with H-28, appendix 10.
6. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
7. All undimensioned pictorial representations are for reference purposes only.
8. Series TNC, socket contact interface in accordance with MIL-STD-348, 313.2.

FIGURE 1. General configurations.

AMSC N/A
FSC 5935

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Inches	mm
.002	0.05
.006	0.15
.010	0.25
.032	0.81
.036	0.91
.077	1.96
.4375	11.11

CATEGORY D

NOTES:
1. Metric equivalents are given for general information only.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Chamfer is optional.
4. Dimensions are in inches.

FIGURE 2. Mating dimensions for socket terminations.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash no. 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) maximum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0101 3/	28-RG058 84-RG223 4/ 128-RG400 60-RG142 5/	A	1.250 (31.75)
0102 3/	29-RG59 4/ 6/ 030-RG062 6/ 90-RG71 6/ 97-RG210 6/		
0103 3/	54-RG122 4/		
0104 3/	111-RG303 4/ 5/		
0117 3/	110-RG302 4/5/6/		
0018 3/	113-RG316 4/ 5/ 119-RG174		
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 7/ SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE			
0010 3/	28-RG058 4/8/ 111-RG303 5/8/	A	1.500 (38.10)
0011 3/	84-RG223 4/8/ 128-RG400 8/ 60-RG142 5/8/		
0012 3/	29-RG59 4/6/9/ 030-RG062 6/9/ 97-RG210 6/9/		
0013 3/	54-RG122 4/10/		
0014 3/	90-RG71 4/6/9/		
0021 3/	110-RG302 4/5/6/ 9/		
0022 3/	113-RG316 4/5/11/ 119-RG174 11/		
CATEGORY D - FIELD REPLACEABLE - DEFINED PIECE PARTS 7/12/13/			
0501 3/	54-RG122 4/	A	1.468 (37.31)
0502 3/	095-RG180 4/ 6/		
0503 3/	60-RG142 4/ 128-RG400		
0504 3/	28-RG058 4/		

1/ For cross-reference of dash number to superseded part number or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

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4/ Cable to be used when performing tests requiring cable except as in 5/ amd 6/.

5/ Cable to be used for the +200°C temperature cycling tests.

6/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

7/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

8/ M22520/5-19 closure B or M22520/5-05 closure A
M22520/5-11 closure A
M22520/5-57 closure A

9/ M22520/5-19 closure A or M22520/5-07 closure A
M22520/5-13 closure A
M22520/5-59 closure A

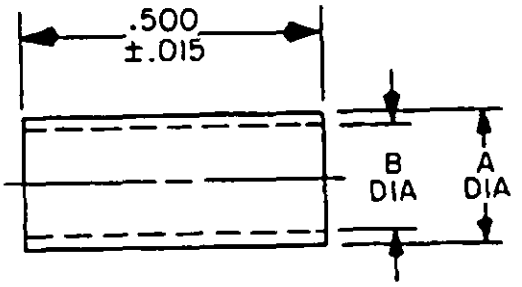
10/ M22520/5-41 closure B or M22520/5-05 closure B
M22520/5-09 closure A

11/ M22520/5-35 closure B or M22520/5-03 closure A

12/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

13/ Not to be used in Army equipment.

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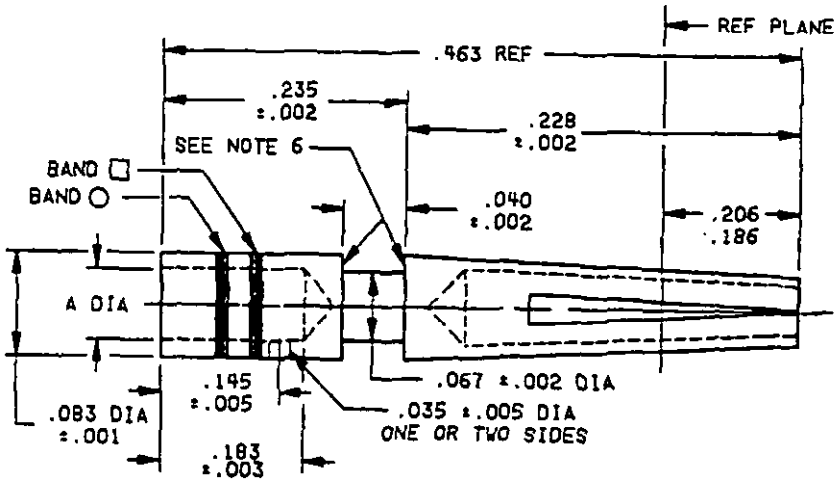
CRIMP FERRULE

Dash No.	Ferrule No. 2/	A ±.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner M22520/5-
0501	29-50	.212	.175	M22520/5-01	05, 41 Closure B or 9 Closure A
0503	29-51	.250	.220		05, 11, 57 Closure A or 19 Closure B
0504	29-52	.245	.206		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

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CENTER CONTACT

Dash no.	Contact no. 2/	A +.001 -.002	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501	29-12	.033	M22520/1-01	M22520/1-12	10 lbs	Orange	Green
0502	29-11	.017			6 lbs	Blue	
0503 0504	29-10	.043	M22520/1-01	M22520/1-12	20 lbs	Red	

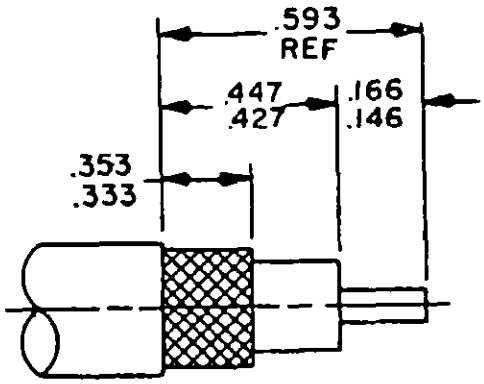
1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.015	0.38
.017	0.43
.033	0.84
.035	0.89
.040	1.02
.043	1.09
.067	1.70
.083	2.11
.145	3.68
.175	4.45
.183	4.65
.186	4.72
.206	5.59
.212	5.38
.220	5.59
.228	5.79
.235	5.97
.245	6.22
.250	6.35
.463	11.76
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.146	3.71
.166	4.22
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.593	15.06

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 4 to 6 inch-pounds.

Mating characteristics:

See MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin - .057 diameter minimum (non-closed entry contacts only).

Insertion depth - .125 minimum.

Number of insertions - 1.

Insertion force test - Steel test pin diameter .054 minimum.

Test pin finish - 16 microinches.

Insertion force - 2 pounds maximum.

Withdrawal force test: Steel test pin diameter .052 maximum.

Withdrawal force - 2 ounces minimum.

Test pin finish - 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

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Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Item 16 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06 + .007 F (F in GHz).

Group B inspection - VSWR shall be less than 1.05 + .005 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.10.

Connector durability: 500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum:

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 1,500 volts rms minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock : Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 375 volts rms, minimum.

Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

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Cable retention force:

Noncrimp assemblies: 40 pounds minimum.

Crimp assemblies:

10 pounds minimum for cables .155-.189 OD.

20 pounds minimum for cables .190-.229 OD.

30 pounds minimum for cables .230-.249 OD.

40 pounds minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:

.18 dB maximum tested at 9 GHz.

.06 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/29- (dash number from table I or "B" number from table III).

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TABLE II. Group Qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
I	27-0101	27-0101
	27-0103	27-0002
	27-0104	27-0103
	27-0018	27-0104
	28-0101	27-0117
	28-0103	27-0018
	28-0104	28-0101
	28-0018	28-0102
	29-0101	28-0103
	29-0103	28-0104
	29-0104	28-0117
	29-0018	28-0018
		29-0101
		29-0102
		29-0103
		29-0104
		29-0017
		29-0018
II	27-0002	27-0002
	27-0117	27-0117
	28-0102	28-0102
	28-0117	28-0117
	29-0102	29-0102
	29-0117	29-0117
III	27B0005	27B0005
	27B0006	27B0006
	27B0008	27B0007
	27B0015	27B0008
	27B0016	27B0009
	27B0020	27B0015
	28B0005	27B0016
	28B0006	27B0019
	28B0008	27B0020
	28B0015	28B0005
	28B0016	28B0006
	28B0020	28B0007
	29B0005	28B0008
	29B0006	28B0009
	29B0008	28B0015
	29B0015	28B0016
	29B0016	28B0019
	29B0020	28B0020
		29B0005
		29B0006
		29B0007
		29B0008
		29B0009
		29B0015
		29B0016
		29B0019
		29B0020

See footnote at end of table.

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TABLE II. Group Qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
IV	M39012/ 27B0007 27B0009 27B0019 28B0007 28B0009 28B0019 29B0007 29B0009 29B0019	M39012/ 27B0007 27B0009 27B0019 28B0007 28B0009 28B0019 29B0007 29B0009 29B0019
V	27-0010 27-0011 27-0013 27-0022 28-0010 28-0011 28-0013 28-0022 29-0010 29-0011 29-0013 29-0022	27-0010 27-0011 27-0012 27-0013 27-0014 27-0021 27-0022 28-0010 28-0011 28-0012 28-0013 28-0014 28-0021 28-0022 29-0010 29-0011 29-0012 29-0013 29-0014 29-0021 29-0022
VI	27-0012 27-0014 27-0021 28-0012 28-0014 28-0021 29-0012 29-0014 29-0021	27-0012 27-0014 27-0021 28-0012 28-0014 28-0021 29-0012 29-0014 29-0021
VII	28-0501 28-0503 28-0504	28-0501 28-0502 28-0503 28-0504
VIII	28-0502	28-0502

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED). 1/

[NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY.]

Part no. 2/ M39012/29B	Applicable cable 3/ M17/	Dimensions	Inches (millimeters) maximum
0005 4/	28-RG058 5/	A	1.500 (38.10)
0006 4/	60-RG142 6/ 5/ 128-RG400		
0007 4/	29-RG59 5/ 7/ 030-RG062 7/ 97-RG210 7/		
0008 4/	54-RG122 5/		
0009 4/	90-RG71 5/ 7/		
0015 4/	111-RG303 5/ 6/		
0016 4/	84-RG223 5/		
0019 4/	110-RG302 5/6/7/		
0020 4/	113-RG316 5/ 6/ 119-RG174		

1/ For maintenance replacements for category B, see table V.

2/ For cross-reference of dash number to superseded part number or type designation, see table IV.

3/ The latest version of each cable shall be applicable.

4/ Inactive for new design.

5/ Cable to be used when performing tests requiring cable except as in 6/ and 7/.

6/ Cable to be used for the +200°C temperature cycling tests.

7/ These are not 50-ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage and insertion loss are not applicable.

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TABLE IV. Cross-reference of part numbers. 1/

Preferred part number	Substitute for part number or type designation 2/3/
M39012/29	M39012/29-
-0101	0001
-0102	0002
-0103	0003
-0104	0004
-0117	0017
B0005	0005
B0006	0006
B0007	0007
B0008	0008
B0009	0009
B0015	0015
B0016	0016
B0019	0019
B0020	0020

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged.

2/ The superseded part number or the type designation is for cross-reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/29-XXXX shall be used in all cases for marking and identifying the connector.

3/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

TABLE V. Maintenance replacements for category B.

Category B no. 1/	Category C dash no.	Category A dash no.	Category D dash no.
B0005	0010	0101	0504
B0006	0011	0101	0503
B0007	0012	0102	---
B0008	0013	0103	0501
B0009	0014	0102	---
B0015	0010	0104	---
B0016	0011	0101	---
B0019	0021	0117	---
B0020	0022	0018	---

1/ Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3599-06)

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